

Learning from Videos

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Why focus on video data?

- ▶ **Ubiquity of video data:** streaming, social media, autonomous systems
- ▶ **Need for temporal understanding:** beyond static images to dynamic events
- ▶ **Generative potential:** create, predict, and simulate future frames
- ▶ **Real-world impact:** surveillance, sports analytics, healthcare, AR/VR, content creation

By the end of this session, you will be able to:

- ▶ Define video data and temporal concepts.
- ▶ Identify challenges in acquiring and processing video.
- ▶ Implement models from single-frame CNNs to spatio-temporal Transformers.
- ▶ Extract motion features and recognize actions.
- ▶ Apply self-supervised video representation learning.
- ▶ Critically assess limitations and propose future research.

Video *generation* and vision–language models (CLIP) are covered later in the course.

Learning from Videos: **Introduction**

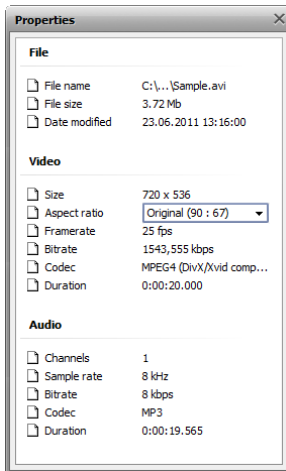
- ▶ **Video** is a sequence of images (frames) captured over time.
- ▶ Each frame is a static image, but the sequence captures motion and temporal dynamics.
- ▶ Videos can be continuous (e.g., surveillance footage) or discrete (e.g., video clips).
- ▶ Key components:
 - Frame rate: number of frames per second (FPS).
 - Resolution: dimensions of each frame.
 - Duration: total length of the video.

What is Video Data? (cont.)

Video are everywhere, and in every field.
Some span hours, while others are just a few seconds long



What is Video Data? (cont.)

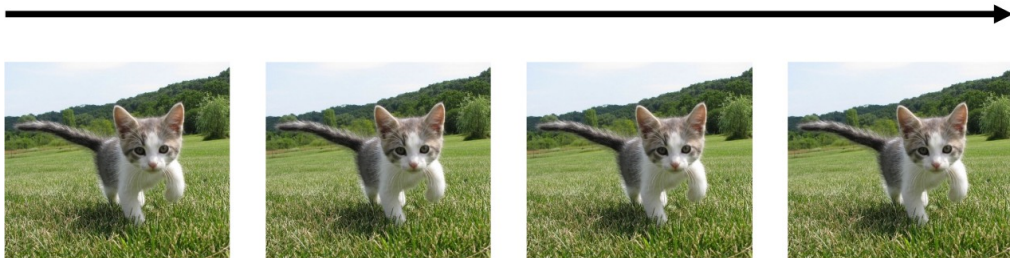


[Source: <https://onlinehelpstaticcontents.avs4you.com/en/images/videoeditor/videoproperties.png>]

What is Video Data? (cont.)

RESOLUTION	AVERAGE DATA USAGE PER HOUR
480P (SD)	360 MB
720P (HD)	900 MB
1080P (FULL HD)	1.5 GB
4K (ULTRA HD)	7.2 GB

A video is a **sequence** of images
4D tensor: $T \times 3 \times H \times W$
(or $3 \times T \times H \times W$)



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 7]

- ▶ Cameras capture video data with various properties that affect quality and usability.
- ▶ Key properties include:
 - **FPS (Frames per Second):** Determines the temporal resolution of the video; higher FPS captures smoother motion.
 - **Spatial Resolution:** Refers to the pixel dimensions of each frame (e.g., 1920x1080); affects image detail.
 - **Field of View (FOV):** The angular extent of the observable scene captured by the camera; wide FOV covers more area.
 - **Depth Sensors:** Devices like RGB-D cameras and LiDAR provide depth information, enabling 3D scene understanding.

Camera Properties (cont.)

30FPS



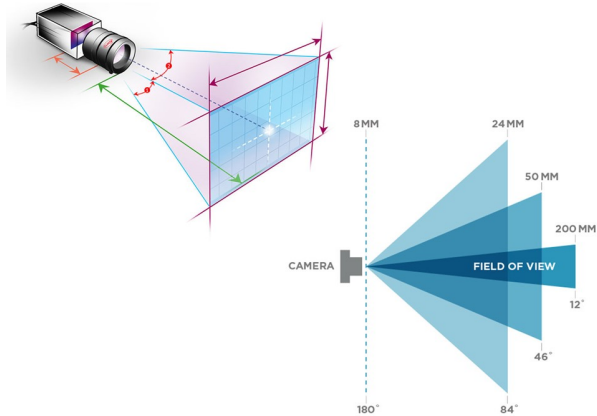
15FPS



1 FPS



[Source: <https://home-cdn.reolink.us/wp-content/uploads/2023/06/200657281687244248.0603.jpg>]

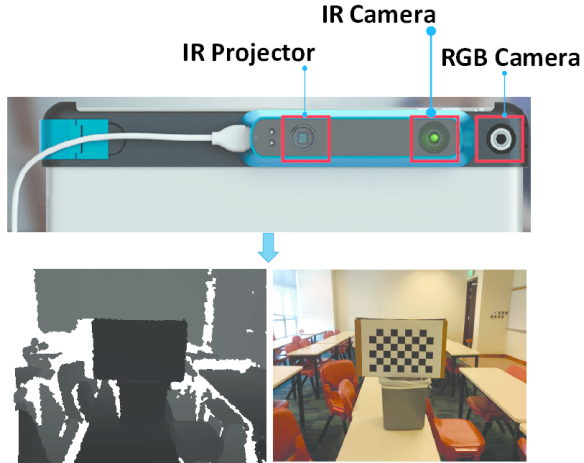


[Source: <https://kintronics.com/wp-content/uploads/2022/08/FOV.jpg>]

Camera Properties (cont.)



[Source: <https://home-cdn.reolink.us/wp-content/uploads/2019/08/analog-vs-ip-security-camera-resolution.jpg>]



[Tang et al., "Enhanced RGB-D Mapping Method for Detailed 3D Indoor and Outdoor Modeling", Sensors 16(10):1589, 2016, Fig. 1]

- ▶ **Single Frame:** Static content; no motion cues.
- ▶ **Short Clip:** Captures motion trajectories.
- ▶ **Temporal Window:** Enables extraction of velocity and acceleration.
- ▶ **Temporal Features:** Techniques such as frame differencing and optical flow capture motion information.

What Does Temporal Mean? (cont.)

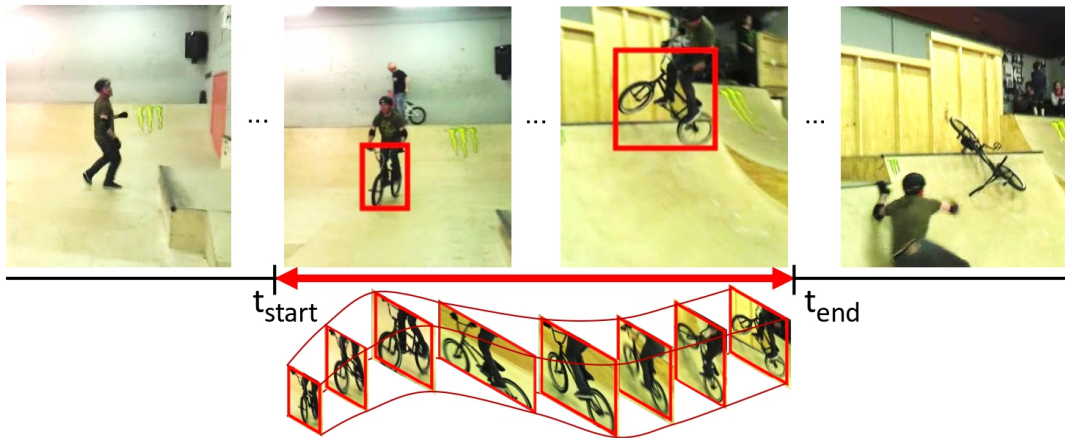


Illustration of temporal features in video: single frame, short clip, and motion extraction.

[Yang et al., "TubeDETR: Spatio-Temporal Video Grounding with Transformers", CVPR 2022, Fig. 1]

Learning from Videos: **Data Acquisition & Challenges**

- ▶ Video data can be acquired from various sources, including:
 - **Cameras:** Digital cameras, smartphones, and webcams.
 - **Drones:** Aerial footage for surveillance or mapping.
 - **Webcams:** Live streaming or recorded video.
 - **Public Datasets:** Pre-recorded videos available for research.
- ▶ Key considerations in data acquisition:
 - Quality of the recording device.
 - Environmental conditions (lighting, weather).
 - Ethical considerations (privacy, consent).

► Modern sources for video data:

- **User-generated content:** Platforms like YouTube and TikTok provide vast amounts of diverse video data.
- **Surveillance & IoT:** Cameras embedded in city infrastructure, public spaces, and smart devices continuously capture video streams.
- **Automotive:** Dashcams and sensors in autonomous vehicles generate large-scale driving and traffic video datasets.

► **Metadata and tags:** Videos are often accompanied by metadata (timestamps, GPS, tags) that facilitate search, retrieval, and organization.

Popular Video Datasets:

- ▶ **UCF101:** 13,000+ video clips covering 101 human action categories.
- ▶ **Kinetics-700:** 650,000 video clips annotated with 700 action classes.
- ▶ **ActivityNet:** Large-scale dataset with untrimmed videos for temporal action localization.
- ▶ **Something-Something v2:** Focuses on fine-grained, object-centric actions.
- ▶ **Ego4D:** Egocentric videos capturing daily activities from a first-person perspective.

Datasets

12,414 machine learning datasets

🔔 Share your dataset with the ML community!

🔍

🔍

Best match

Filter by Modality (clear)

Videos

Texts 3332

Images 3274

Audio 484

Medical 424

Other 399

Filter by Task

Action Recognition 80

Video Understanding 44

Object Tracking 40

Object Detection 39

Video Question Answering 38

1021 dataset results for Videos



MS COCO (Microsoft Common Objects in Context)

The MS COCO (Microsoft Common Objects in Context) dataset is a large-scale object detection, segmentation, key-point detection, and captioning dataset. The dataset consists of 328K...

11,848 PAPERS • 96 BENCHMARKS



UCF101 (UCF101 Human Actions dataset)

UCF101 dataset is an extension of UCF50 and consists of 13,320 video clips, which are classified into 101 categories. These 101 categories can be classified into 5 types (Body motion, Human...

1,853 PAPERS • 27 BENCHMARKS



Kinetics (Kinetics Human Action Video Dataset)

The Kinetics dataset is a large-scale, high-quality dataset for human action recognition in videos. The dataset consists of around 500,000 video clips covering 600 human action classes...

1,337 PAPERS • 31 BENCHMARKS



HMDB51

The HMDB51 dataset is a large collection of realistic videos from various sources, including movies and web videos. The dataset is composed of 6,766 video clips from 51 action categories...

837 PAPERS • 12 BENCHMARKS



ActivityNet

The ActivityNet dataset contains 200 different types of activities and a total of 849 hours of videos collected from YouTube. ActivityNet is the largest benchmark for temporal activity de...

803 PAPERS • 20 BENCHMARKS



Human3.6M

The Human3.6M dataset is one of the largest motion capture datasets, which consists of 3.6

[Source: https://huggingface.co/datasets?task_categories=task_categories:video-classification]

► Issues Working with Videos:

- **High Storage:** Video files are large, requiring significant storage and bandwidth.
- **Variable FPS & Resolution:** Differences in frame rates and resolutions increase preprocessing complexity.
- **Annotation Cost:** Frame-level labeling is time-consuming and expensive.
- **Data Imbalance:** Rare events or classes are often underrepresented in datasets.

Videos are ~30 frames per second (fps)

Size of uncompressed video
(3 bytes per pixel):

SD (640 x 480): **~1.5 GB per minute**

HD (1920 x 1080): **~10 GB per minute**



Input video:
 $T \times 3 \times H \times W$

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 10]



Input video:
 $T \times 3 \times H \times W$

Videos are ~ 30 frames per second (fps)

Size of uncompressed video
(3 bytes per pixel):

SD (640 x 480): **~ 1.5 GB per minute**

HD (1920 x 1080): **~ 10 GB per minute**

Solution: Train on short **clips**: low
fps and low spatial resolution
e.g. $T = 16$, $H=W=112$
(3.2 seconds at 5 fps, 588 KB)

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 11]

► Solutions for Video Data:

- **Compression & Sampling:** Techniques like keyframe extraction and frame sampling help reduce storage and computational requirements.
- **Data Augmentation:** Methods such as frame dropout and temporal jittering increase data diversity and robustness.
- **Synthetic Data:** Simulated environments (e.g., CARLA, AirSim) generate realistic video data for training and testing.
- **Weak & Self-Supervision:** Using unlabeled or partially labeled data reduces reliance on expensive manual annotations.

Learning from Videos: **Video Understanding Models**

- ▶ Video understanding involves analyzing and interpreting video data to extract meaningful information.
- ▶ Key tasks include:
 - **Action recognition:** Identifying actions or activities in videos.
 - **Object detection:** Locating and classifying objects within video frames.
 - **Scene understanding:** Analyzing the context and environment in which actions occur.
 - **Temporal analysis:** Understanding how actions evolve over time.



Input video:
 $T \times 3 \times H \times W$



Swimming
Running
Jumping
Eating
Standing

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 8]

- ▶ To understand videos, models need to look at both:
 - What's in each frame (spatial info).
 - How things move or change (temporal info).
- ▶ How do we do this?
 - Use CNNs to spot patterns in images.
 - Use RNNs or LSTMs to follow changes over time.
 - Use 3D CNNs to look at several frames together.
 - Use Transformers to connect information across the whole video.



Images: Recognize **objects**



Dog
Cat
Fish
Truck



Videos: Recognize **actions**



Swimming
Running
Jumping
Eating
Standing

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 9]

- ▶ These models learn from lots of labeled videos.
- ▶ We check how well they work using accuracy, precision, recall, and F1-score.
- ▶ Where is this useful?
 - Security and surveillance.
 - Self-driving cars.
 - Sports analysis.
 - Healthcare monitoring.
 - Recommending videos online.

- ▶ **Clip Length Selection:** Typically, short clips of 16–64 frames are used to balance computational cost and temporal coverage.
- ▶ **Batch Size vs. Temporal Context:** Increasing clip length provides more temporal context but reduces the possible batch size due to memory constraints.
- ▶ **Pretraining for Transfer Learning:** Models are often pretrained on large-scale video datasets (e.g., Kinetics, Sports-1M) to improve performance on downstream tasks.

Training on Clips: Key Considerations (cont.)

Raw video: Long, high FPS



Training: Train model to classify short **clips** with low FPS



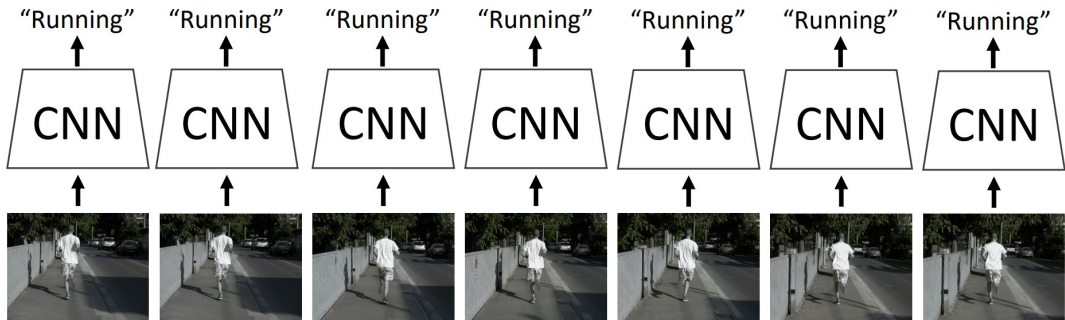
Testing: Run model on different clips, average predictions



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 14]

- ▶ Treat each frame independently
- ▶ Use pretrained ImageNet backbones
- ▶ Ignores temporal context; serves as a baseline

Single-Frame CNN (cont.)



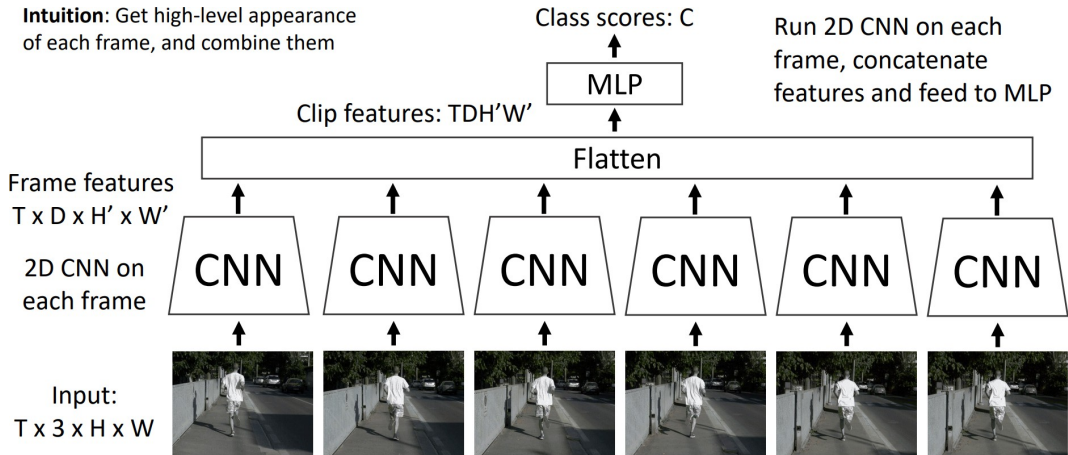
[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 15]

- ▶ **Process frames separately**, then combine their representations.
- ▶ **Fully Connected Fusion**: Concatenate frame features, then use fully connected (FC) layers.
- ▶ **Pooling Fusion**: Apply average or max pooling across time dimension.

Pros: Simple and easy to implement.

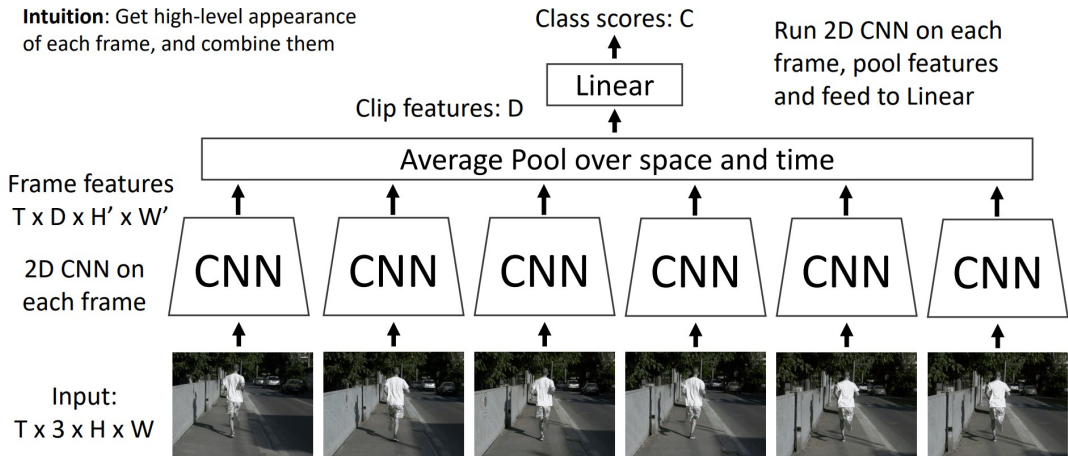
Cons: Weak temporal modeling; ignores complex temporal dependencies.

Late Fusion (with FC layers)



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 16]

Late Fusion (with pooling)

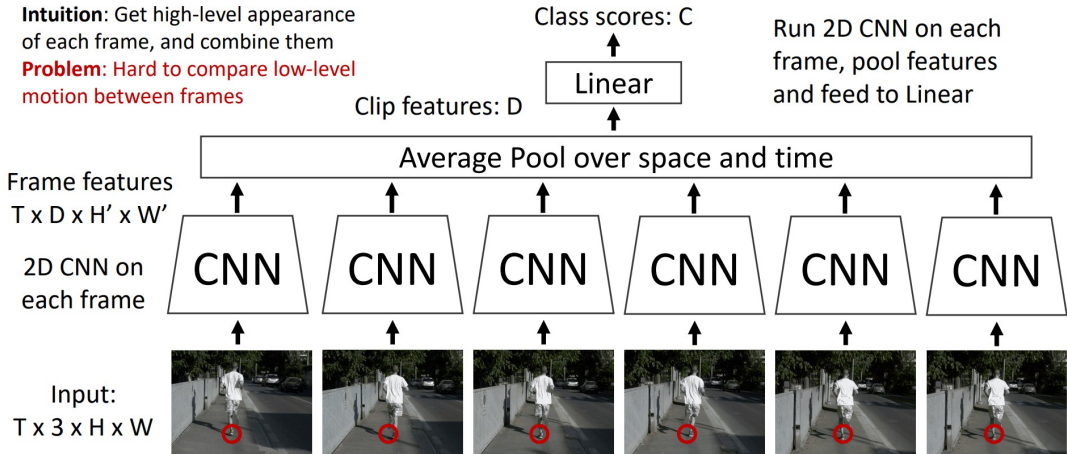


[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 17]

Late Fusion (with pooling) (cont.)

Intuition: Get high-level appearance of each frame, and combine them

Problem: Hard to compare low-level motion between frames

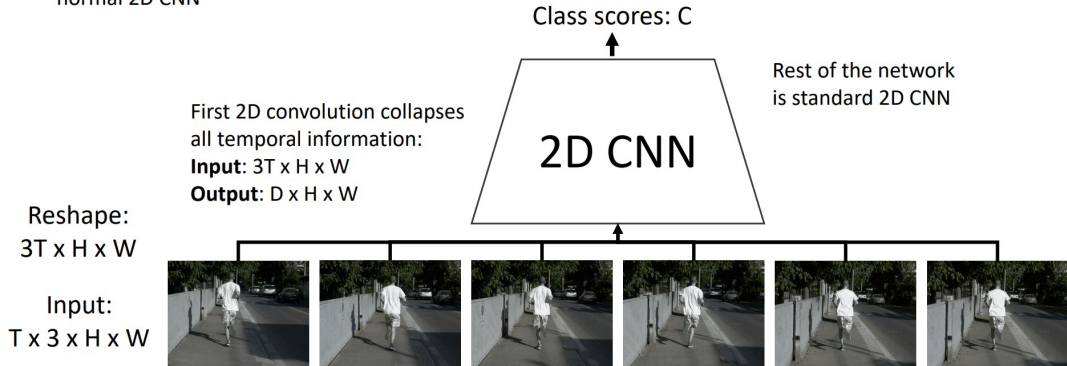


[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 18]

- ▶ **Stack frames as input channels to 2D CNN**
- ▶ Learns short-term motion patterns
- ▶ Limited receptive field along time

Early Fusion: Key Ideas (cont.)

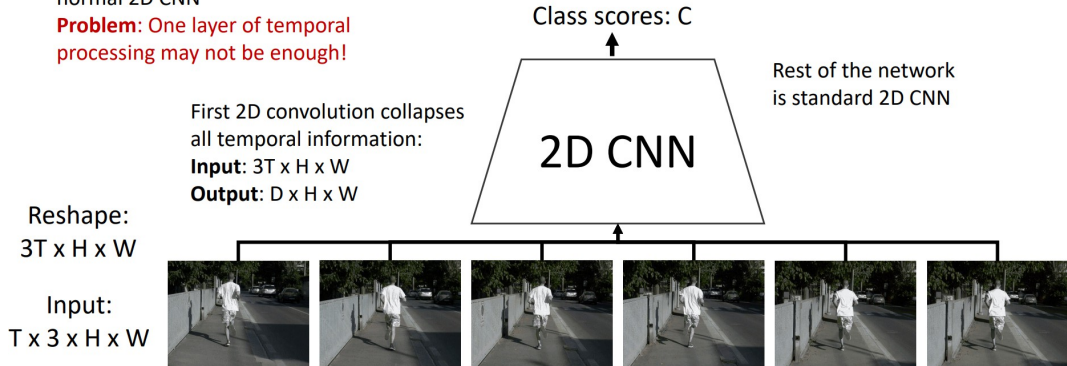
Intuition: Compare frames with very first conv layer, after that normal 2D CNN



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 19]

Intuition: Compare frames with very first conv layer, after that normal 2D CNN

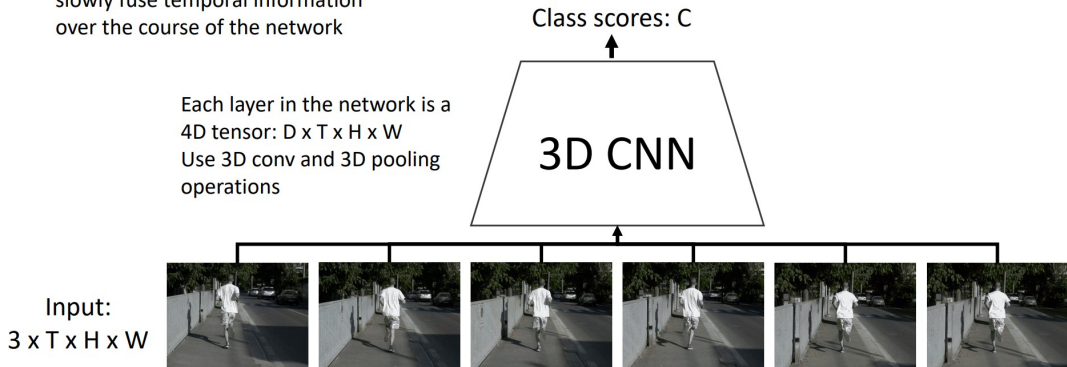
Problem: One layer of temporal processing may not be enough!



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 20]

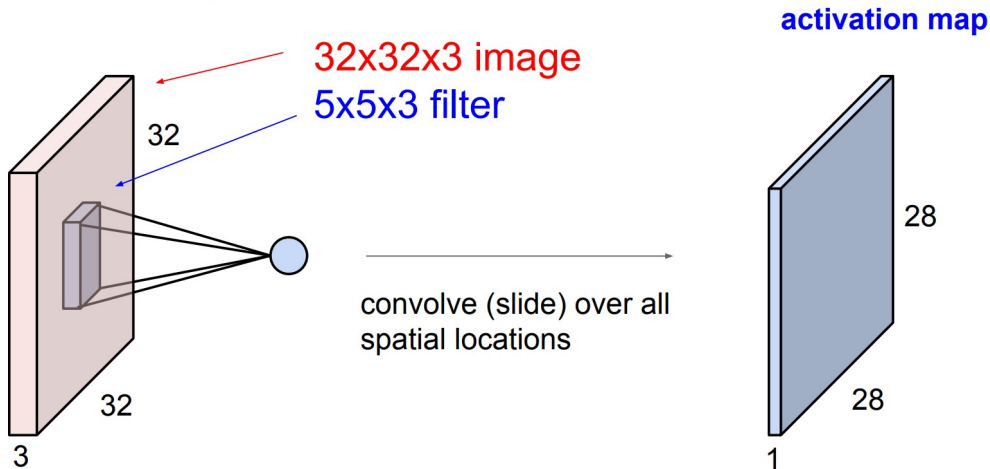
- ▶ **3D Convolutional Networks:** Apply convolutional kernels across both spatial and temporal dimensions.
- ▶ **3D Kernels over Space-Time:** Filters operate on width, height, and time, enabling motion modeling.
- ▶ **Capture Local Spatio-Temporal Features:** Learn patterns that span multiple frames, such as movement or actions.
- ▶ **High Computational Cost:** Increased parameters and operations compared to 2D CNNs, leading to higher resource requirements.

Intuition: Use 3D versions of convolution and pooling to slowly fuse temporal information over the course of the network



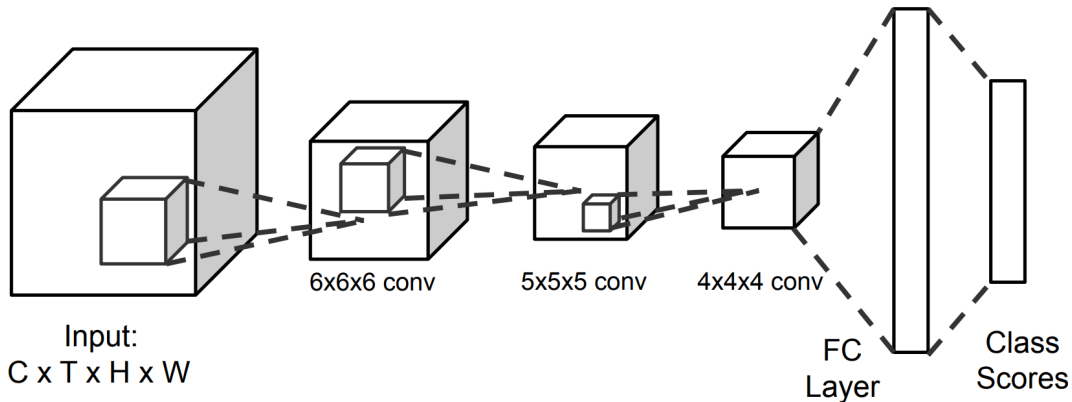
[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 21]

3D CNN: Key Ideas (cont.)



[Source: Stanford CS231n (2023), Lecture 5: Convolutional Neural Networks, slide 56]

3D CNN: Key Ideas (cont.)



[Source: UMich EECS 498 (2019), Lecture 17: 3D Vision, "Processing Voxel Inputs: 3D Convolution", relabelled for video; figure after Wu et al., "3D ShapeNets: A Deep Representation for Volumetric Shapes", CVPR 2015]

Introduced by Tran et al., ICCV 2015

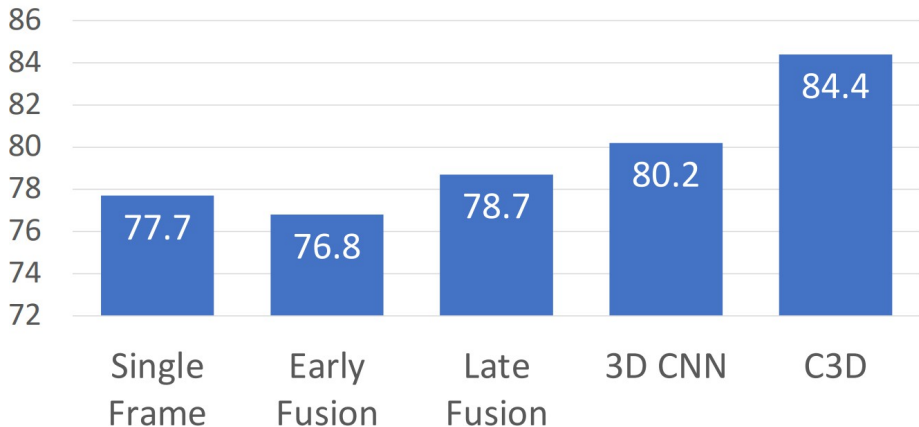
- ▶ Uses $3 \times 3 \times 3$ convolutions on 16-frame video clips.
- ▶ Demonstrated strong performance on the UCF101 action recognition benchmark.
- ▶ **3D Convolutional Networks:** Apply 3D convolutional kernels to capture spatio-temporal features.
- ▶ **Temporal Context:** C3D captures motion patterns by considering multiple frames simultaneously.
- ▶ **Pooling and Normalization:** Use pooling layers to reduce dimensionality and normalize features.
- ▶ **End-to-End Training:** Trained on large video datasets for tasks like action recognition.

- ▶ 3D CNN that uses all $3 \times 3 \times 3$ convolutions and $2 \times 2 \times 2$ pooling (except Pool1, which is $1 \times 2 \times 2$).
- ▶ Released model pretrained on Sports-1M: widely used as a video feature extractor.
- ▶ **Problem:** $3 \times 3 \times 3$ convolutions are very expensive!
 - AlexNet: 0.7 GFLOP
 - VGG-16: 13.6 GFLOP
 - C3D: 39.5 GFLOP ($2.9 \times$ VGG!)

Layer	Size
Input	$3 \times 16 \times 112 \times 112$
Conv1 ($3 \times 3 \times 3$)	$64 \times 16 \times 112 \times 112$
Pool1 ($1 \times 2 \times 2$)	$64 \times 16 \times 56 \times 56$
Conv2 ($3 \times 3 \times 3$)	$128 \times 16 \times 56 \times 56$
Pool2 ($2 \times 2 \times 2$)	$128 \times 8 \times 28 \times 28$
Conv3a ($3 \times 3 \times 3$)	$256 \times 8 \times 28 \times 28$
Conv3b ($3 \times 3 \times 3$)	$256 \times 8 \times 28 \times 28$
Pool3 ($2 \times 2 \times 2$)	$256 \times 4 \times 14 \times 14$
Conv4a ($3 \times 3 \times 3$)	$512 \times 4 \times 14 \times 14$
Conv4b ($3 \times 3 \times 3$)	$512 \times 4 \times 14 \times 14$
Pool4 ($2 \times 2 \times 2$)	$512 \times 2 \times 7 \times 7$
Conv5a ($3 \times 3 \times 3$)	$512 \times 2 \times 7 \times 7$
Conv5b ($3 \times 3 \times 3$)	$512 \times 2 \times 7 \times 7$
Pool5	$512 \times 1 \times 3 \times 3$
FC6	4096
FC7	4096
FC8	C

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 38]

Sports-1M Top-5 Accuracy



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 40]

Learning from Videos: **Motion & Temporal Dynamics**

- ▶ Motion as a primary cue for recognising actions in videos.
- ▶ Importance of separating the analysis of appearance and motion.
- ▶ Motion patterns often provide more discriminative information than static appearance.

- ▶ Computes a dense motion field between consecutive frames.
- ▶ Popular algorithms:
 - Farneback
 - TV-L1
- ▶ Optical flow is commonly used as input to Two-Stream Networks for action recognition.

- ▶ **Dense Motion Field:** Represents motion vectors for every pixel between two frames.
- ▶ **Temporal Coherence:** Assumes that motion is smooth and continuous over time.
- ▶ **Applications:** Used in action recognition, video segmentation, and object tracking.

Optical Flow: Key Concepts (cont.)

Image at frame t

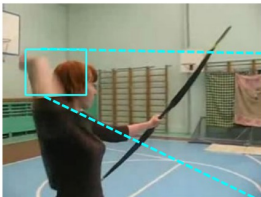
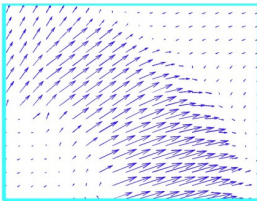


Image at frame $t+1$

Optical flow gives a displacement field F between images I_t and I_{t+1}

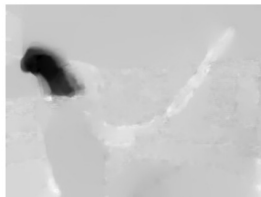


Tells where each pixel will move in the next frame:

$$F(x, y) = (dx, dy)$$

$$I_{t+1}(x+dx, y+dy) = I_t(x, y)$$

Horizontal flow dx

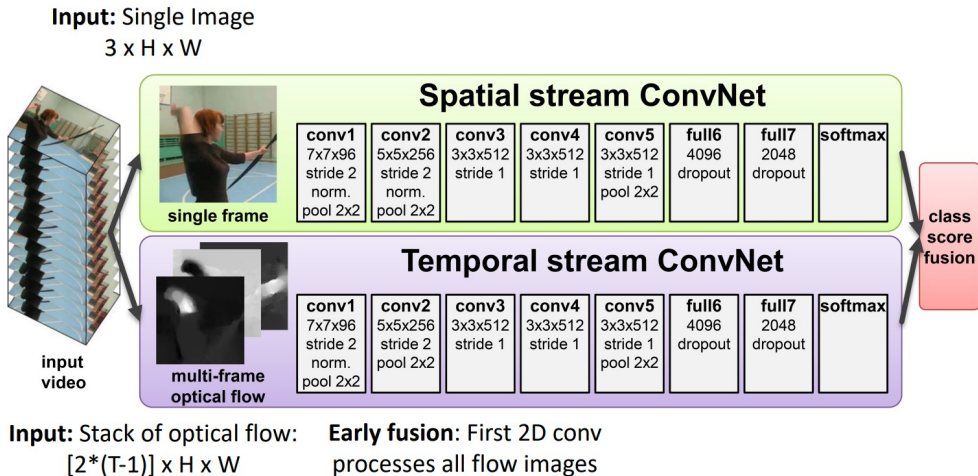


Vertical Flow dy

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 44]

- ▶ **Two-Stream Network:** Processes both RGB (appearance) and optical flow (motion) streams.
- ▶ Streams can be fused at various depths:
 - *Early fusion:* Combine features at initial layers.
 - *Late fusion:* Combine predictions or high-level features.
- ▶ Improves robustness to background and enhances motion understanding.

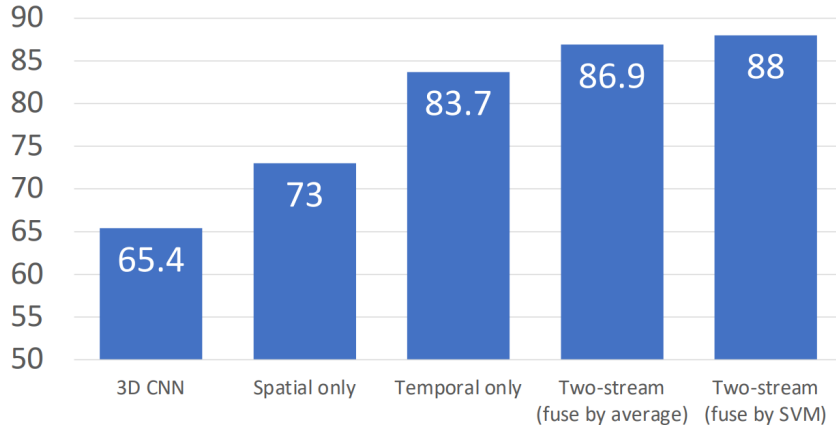
Two-Stream Network Architecture



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 45]

- ▶ **RGB Stream:** Standard CNN (e.g., ResNet) for appearance features.
- ▶ **Optical Flow Stream:** Separate CNN for motion features.
- ▶ **Fusion:** Can be done at different stages:
 - Early: Concatenate features from both streams.
 - Late: Combine predictions from both streams.

Accuracy on UCF-101

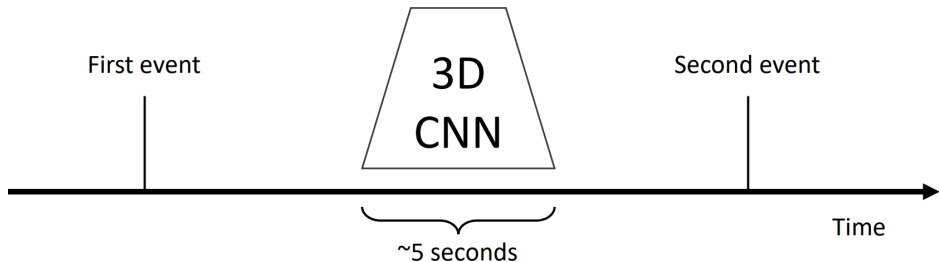


[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 46]

Understanding complex actions in videos often requires analyzing long-term temporal structure, rather than relying solely on short clips. This involves capturing dependencies and patterns that span extended periods, which are crucial for recognizing actions composed of multiple stages or events.

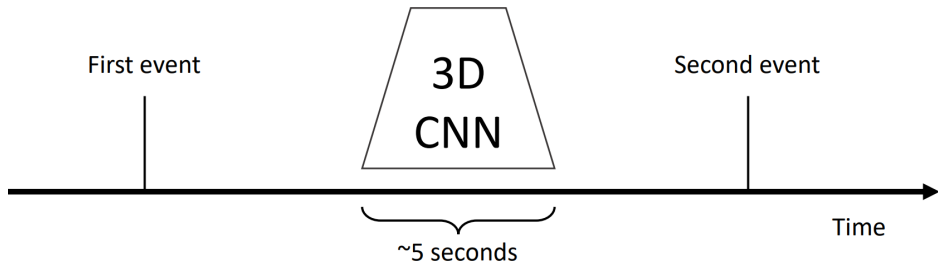
- ▶ **Temporal Segmentation:** Divides a video into meaningful segments based on changes in activity.
- ▶ **Temporal Action Proposals:** Identifies candidate time intervals where actions may occur.

- ▶ So far, all our temporal CNNs only model local motion between frames in very short clips of approximately 2–5 seconds.
- ▶ What about long-term structure?



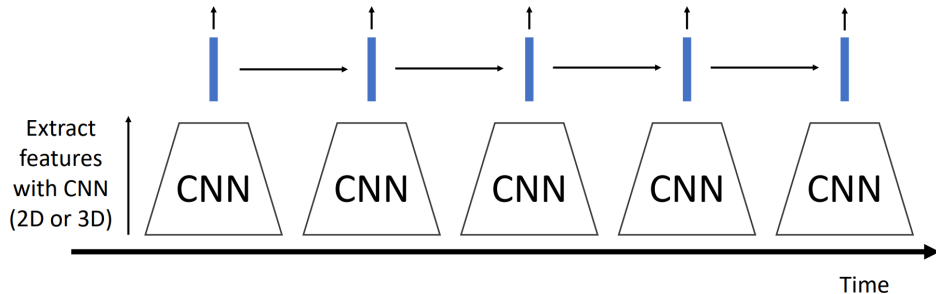
[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 47]

- ▶ So far, all our temporal CNNs only model local motion between frames in very short clips of approximately 2–5 seconds.
- ▶ What about long-term structure?
- ▶ **We know how to handle sequences!**
- ▶ How about **recurrent networks**?



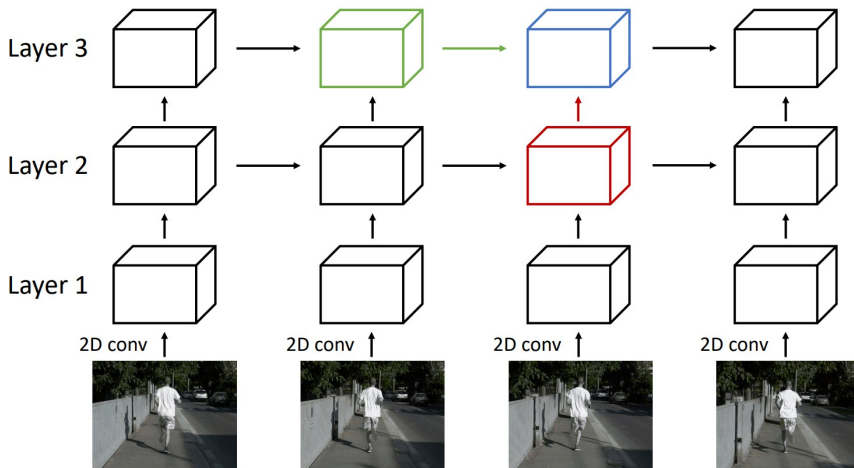
[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 47]

- ▶ Process local features using recurrent network (e.g. LSTM)
- ▶ Many to many: one output per video frame



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 52]

Recurrent Convolutional Network



Entire network
uses 2D feature
maps: $C \times H \times W$

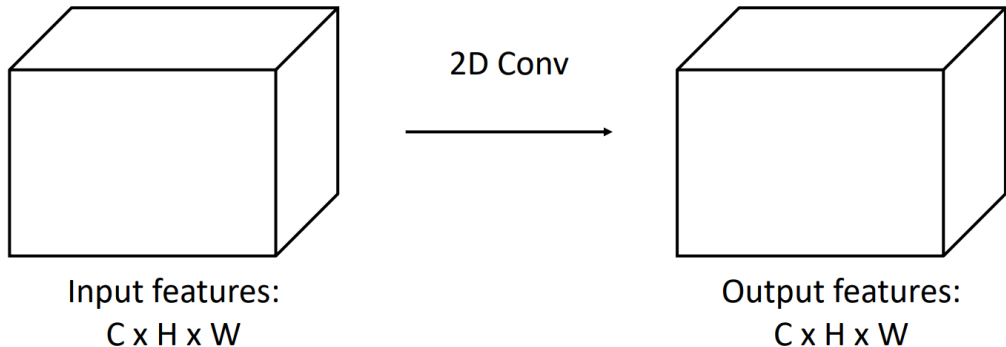
Each depends on
two inputs:

1. Same layer,
previous timestep
2. Prev layer, same
timestep

Use different
weights at each
layer, share weights
across time

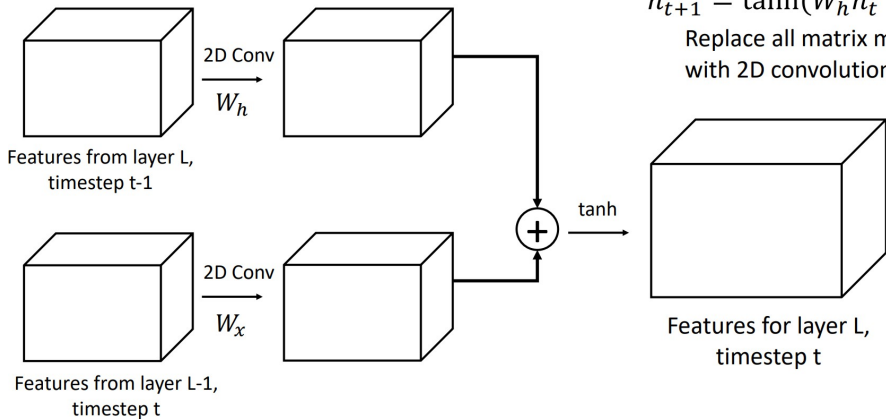
Ballas et al, "Delving Deeper into
Convolutional Networks for Learning
Video Representations", ICLR 2016

Normal 2D CNN:



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 58]

Recurrent Convolutional Network (cont.)



Recall: Vanilla RNN

$$h_{t+1} = \tanh(W_h h_t + W_x x)$$

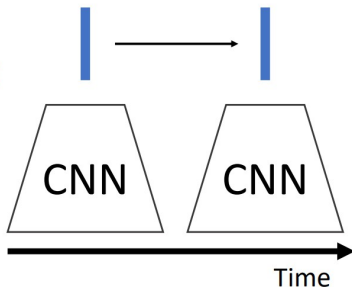
Replace all matrix multiply
with 2D convolution!

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 60]

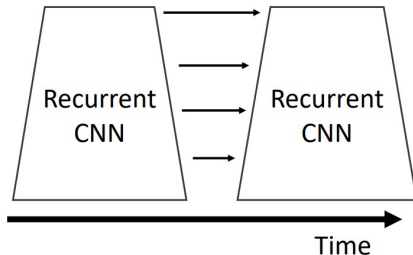
Modelling Long-Term Structure: RNNs vs. CNNs

RNN: Infinite
temporal extent
(fully-connected)

CNN: finite
temporal extent
(convolutional)



Recurrent CNN: Infinite
temporal extent
(convolutional)

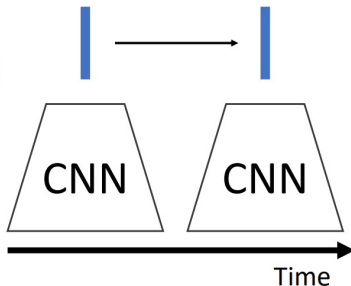


[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 62]

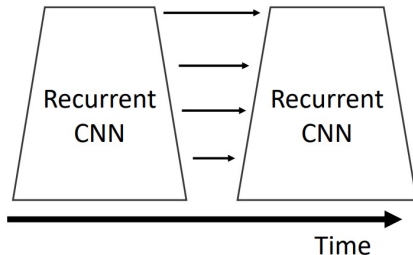
Problem: RNNs are slow for long sequences (can't be parallelized)

RNN: Infinite temporal extent (fully-connected)

CNN: finite temporal extent (convolutional)



Recurrent CNN: Infinite temporal extent (convolutional)



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 63]

Several approaches have been developed to effectively model long-term temporal dependencies in videos:

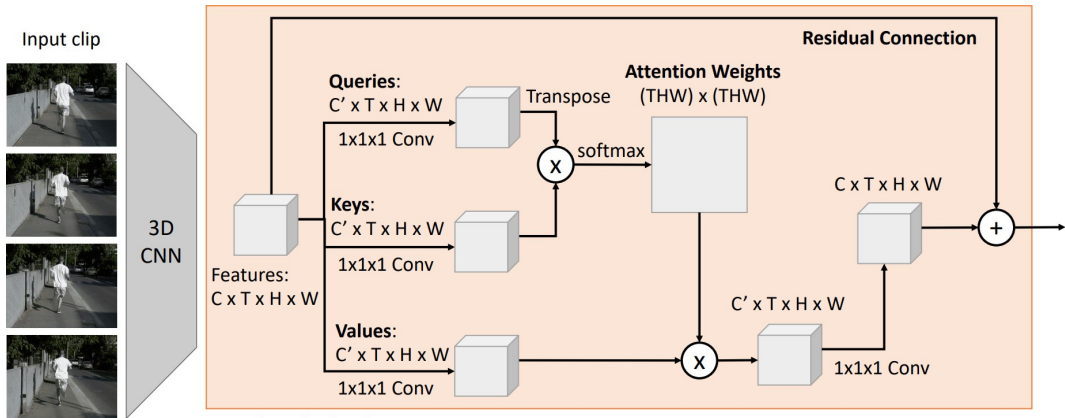
- ▶ **Hierarchical RNNs and LSTMs:** Stack multiple recurrent layers to capture both short-term and long-term dependencies at different temporal scales.
- ▶ **Temporal Segment Networks (TSN):** Sample frames or segments sparsely across the entire video and aggregate their representations to model long-range structure.
- ▶ **Memory Networks:** Utilize external memory modules to store and retrieve information across hundreds of frames, enabling reasoning over extended temporal contexts.

Spatio-Temporal Self-Attention extends the Transformer architecture to video data by modeling dependencies across both spatial and temporal dimensions.

- ▶ **Transformers applied to video:** Video frames are treated as sequences, enabling the use of self-attention to capture relationships across space and time.
- ▶ **Factorized Attention (e.g., TimeSformer):** Attention is applied separately along spatial and temporal axes, reducing computational cost compared to full spatio-temporal attention.
- ▶ **Benefits:** Enables modeling of global context and long-range dependencies in videos.
- ▶ **Costs:** Self-attention has quadratic complexity with respect to sequence length, making it computationally expensive for long videos.

- ▶ **Self-Attention Mechanism:** Captures long-range dependencies in both spatial and temporal dimensions.
- ▶ **Multi-Head Attention:** Allows the model to focus on different parts of the input sequence simultaneously.

Spatio-Temporal Self-Attention (Nonlocal Block)



Nonlocal Block

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 72]

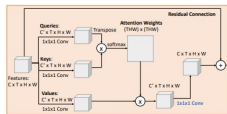
Spatio-Temporal Self-Attention (Nonlocal Block) (cont.)

Input clip



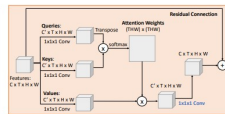
3D CNN

We can add nonlocal blocks into existing 3D CNN architectures.
But what is the best 3D CNN architecture?



Nonlocal Block

3D CNN



Nonlocal Block

3D CNN

Running

[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 74]

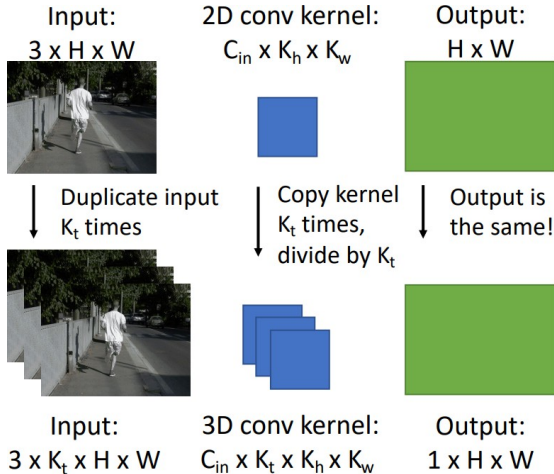
- ▶ There has been a lot of work on architectures for images.
- ▶ Can we reuse image architectures for video?
- ▶ Idea: take a 2D CNN architecture.
- ▶ Replace each 2D $K_h \times K_w$ conv/pool layer with a 3D $K_t \times K_h \times K_w$ version.
- ▶ Can use weights of 2D conv to initialize 3D conv: copy K_t times along time and divide by K_t .
- ▶ This gives the same result as 2D conv given “constant” video input.

► Introduced by Carreira and Zisserman, CVPR 2017

► Key Ideas:

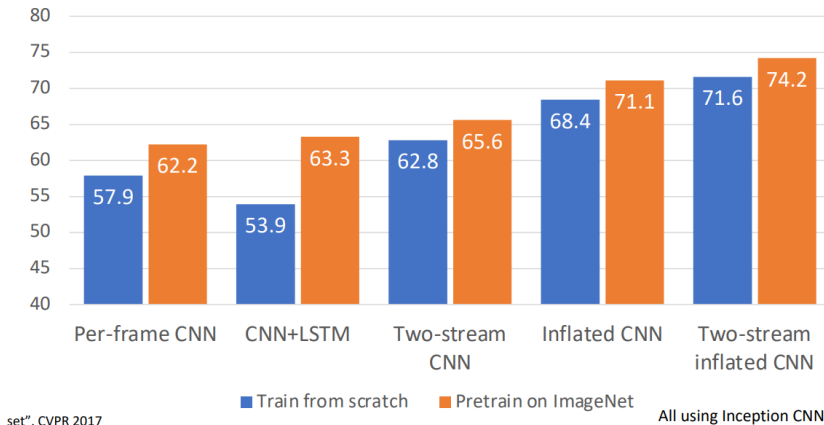
- **Inflation of 2D Filters:** Converts 2D convolutional filters into 3D by replicating them across time.
- **Temporal Modeling:** Captures motion information effectively by reusing pre-trained 2D networks.
- **End-to-End Training:** Trained on large video datasets, allowing transfer learning to new tasks.
- **Performance Boost:** Significantly improves action recognition accuracy compared to previous methods.

Inflating 2D Networks to 3D (I3D) (cont.)



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 78]

Top-1 Accuracy on Kinetics-400



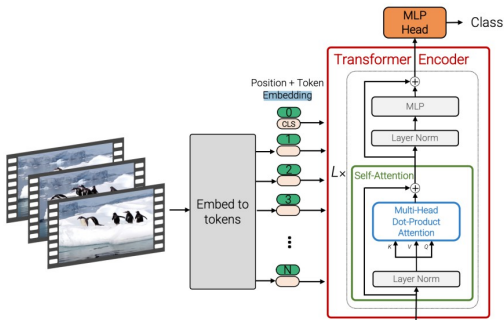
[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 79]

Vision Transformers (ViTs) extend to video by treating a clip as a sequence of tokens, so self-attention can relate patches across both space and time.

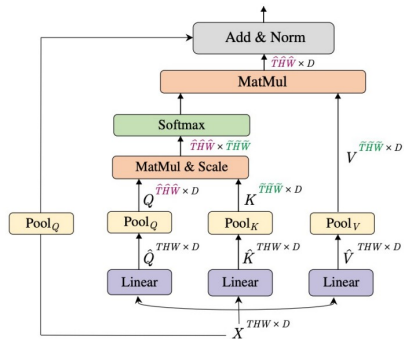
- ▶ **Patch Embedding:** Each frame is divided into patches that are linearly embedded; ViViT's *tubelet embedding* instead embeds spatio-temporal tubes spanning several frames, so each token already carries motion information.
- ▶ **Temporal Encoding:** Positional encodings are extended to the time dimension to preserve frame order.
- ▶ **Factorised Variants:** ViViT factorises the encoder into spatial then temporal stages, and pooling-based designs such as MViT progressively reduce the number of tokens, cutting the cost of full spatio-temporal attention.

[Arnab et al., "ViViT: A Video Vision Transformer," ICCV 2021]

Factorized attention: Attend over space / time



Pooling module: Reduce number of tokens



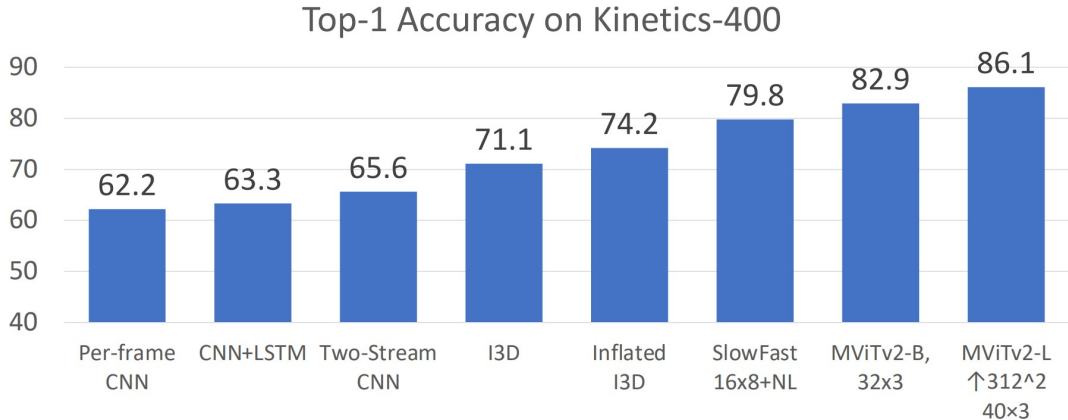
Bertasius et al, "Is Space-Time Attention All You Need for Video Understanding?", ICML 2021

Arnab et al, "ViViT: A Video Vision Transformer", ICCV 2021

Neimark et al, "Video Transformer Network", ICCV 2021

Fan et al, "Multiscale Vision Transformers", ICCV 2021

Li et al, "MViTv2: Improved Multiscale Vision Transformers for Classification and Detection", CVPR 2022



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 81]

Advanced Video Understanding

Problem Statement:

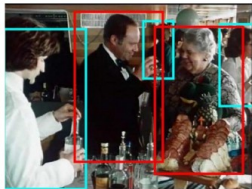
Given a long untrimmed video, the goal is to detect all people in both space and time, and classify the activities they are performing.

- ▶ **Input:** Untrimmed video sequence.
- ▶ **Output:** For each person, a spatio-temporal tube (bounding boxes across frames) and an activity label.

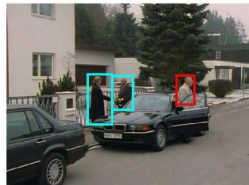
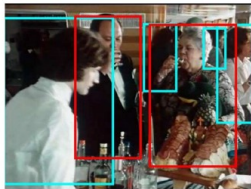
Example: AVA Dataset

The AVA (Atomic Visual Actions) dataset annotates 15-minute movie clips with person bounding boxes and labels from a vocabulary of 80 atomic actions, on keyframes sampled at 1 Hz (Gu et al., CVPR 2018).

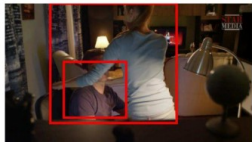
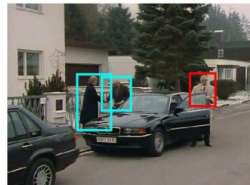
Spatio-Temporal Detection (cont.)



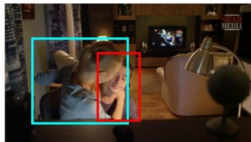
clink glass → drink



open → close



grab (a person) → hug



look at phone → answer phone



[Source: UMich EECS 498 (2022), Lecture 24: Videos, slide 89]

Ego4D is a large-scale dataset designed to advance the understanding of egocentric (first-person) video. It consists of over 3,670 hours of egocentric video footage captured using head-mounted cameras.

- ▶ **Long Videos:** Each video ranges from *1 to 10 hours* in length.
- ▶ **Diversity:** Data collected by *14 teams* across *9 countries*, *74 locations*, involving *931 unique camera wearers*.
- ▶ **Natural-Language Narrations:** Contains *3.85 million* sentences of narrations.
- ▶ **Rich Annotations:** Supports *5 different tasks*:
 - Episodic Memory
 - Hands and Objects
 - Audio-Visual Diarization
 - Social Interactions
 - Forecasting

Key Features of Ego4D:

- ▶ **First-Person Perspective:** Captures videos from the viewpoint of the person wearing the camera, providing a unique perspective on human activities.
- ▶ **Large Scale:** 3,670 hours of video data, making it one of the largest datasets for egocentric video understanding.
- ▶ **Diverse Activities and Participants:** Wide range of activities and a large, varied set of camera wearers, reflecting real-world scenarios.
- ▶ **Comprehensive Annotations:** Natural-language narrations and support for multiple tasks enable comprehensive model training and evaluation.

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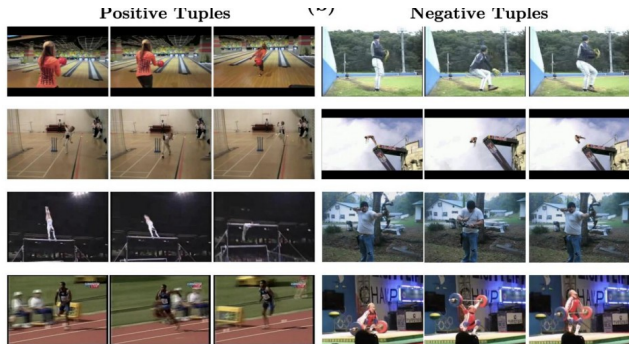
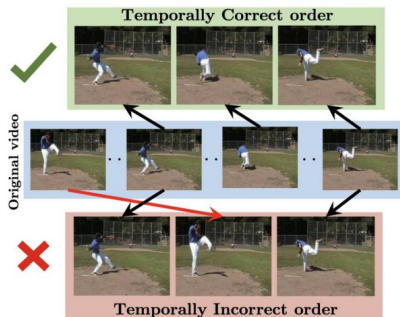


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- ▶ **Temporal Order:** Learning to predict or verify the correct temporal sequence of video frames as a self-supervised task.
- ▶ **Learning Correspondence:** Matching patches across frames, learned from unlabeled video, as a signal for tracking and segmentation.
- ▶ **Cycle Consistency:** Enforcing that matches made forward in time can be traced back to their starting point, ensuring consistency in learned representations.
- ▶ **Video Speedup:** Training models to recognize videos played at different speeds, encouraging temporal understanding.

Training data: Shuffled video frames, original video frames

Pretext task: Predict if the frames are in the correct temporal order (binary classification task).

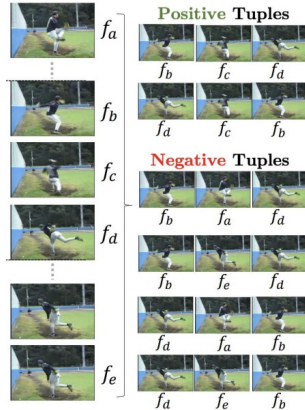


[Misra et al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016]

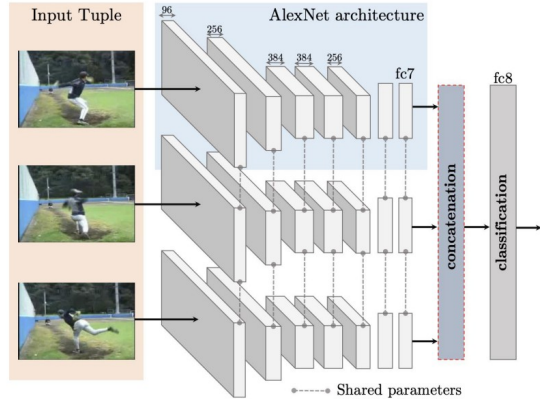
Objective: Learn temporal dependencies and motion patterns in videos.

- ▶ Given a sequence of video frames, randomly shuffle the order of frames.
- ▶ The model receives either the original or shuffled sequence.
- ▶ The objective is to classify whether the input sequence is temporally correct or not.
- ▶ This encourages the model to learn temporal dependencies and motion patterns in videos.

Frame Reordering (cont.)



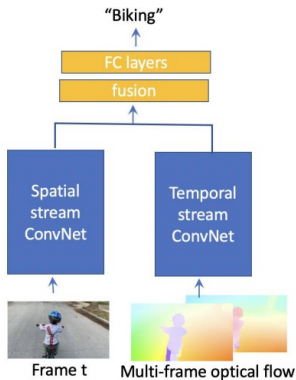
Generating positive and negative examples



Triplet Siamese network for sequence verification

[Misra et al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016]

Transfer Learning: Fine-tune spatial stream for video classification.



Dataset	Initialization	Mean Accuracy
UCF101	Random	38.6
	(Ours) Tuple verification	50.2
HMDB51	Random	13.3
	UCF Supervised	15.2
	(Ours) Tuple verification	18.1

[Misra et al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016]

Benefits:

- ▶ No need for labeled data, as the task is self-supervised.
- ▶ Helps the model learn temporal dynamics and motion patterns in videos.
- ▶ Can be used as a pre-training step before fine-tuning on specific video tasks.

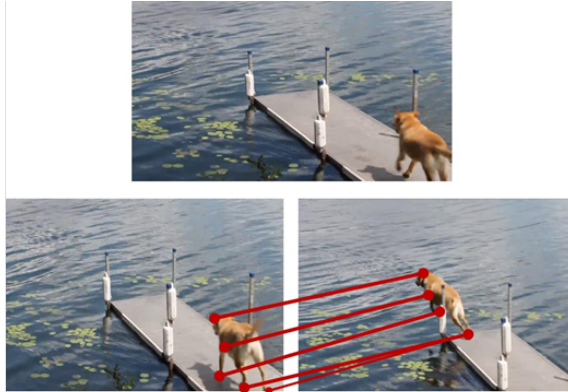
Training data: Unlabeled video

Pretext task: Track a patch backward in time and then forward again; failing to return to the starting point is the training loss (the cycle-consistency of time).

Learning Correspondence:

- ▶ Match patches across frames using the learned features
- ▶ Applications: tracking, video object segmentation, correspondence learning

Ultimate Goal: Correspondence

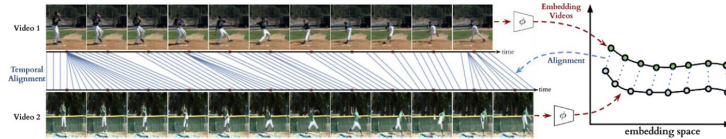


[Wang et al., Learning Correspondence from the Cycle-Consistency of Time, CVPR 2019]

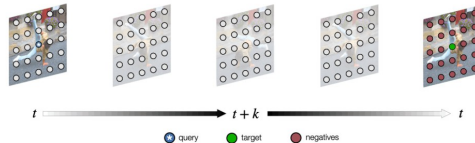
Temporal Cycle Consistency is a self-supervised learning approach that uses the temporal structure of video itself as supervision: correct correspondences form cycles that return to where they started.

- ▶ **Cycle Consistency:** Match a patch or frame forward to another point in time and back again; the training loss penalizes cycles that fail to return to the starting frame.
- ▶ **Across Videos:** The same idea aligns two videos of the same action, matching each frame to its nearest neighbour in the other video's embedding space and back.
- ▶ **Applications:** Video alignment, tracking, and correspondence learning.

Temporal cycle consistency (cont.)



Dwibedi et. al. Temporal Cycle-Consistency Learning, CVPR'19



Jabri et. al, Space time correspondence as Contrastive Random Walk, NeurIPS 2020

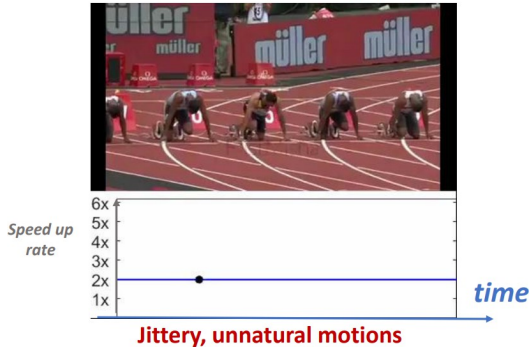
[Dwibedi et al., Temporal Cycle-Consistency Learning, CVPR 2019; Jabri et al., Space-Time Correspondence as a Contrastive Random Walk, NeurIPS 2020]

Learning Speediness is a self-supervised learning approach that focuses on understanding the temporal dynamics of videos by learning to recognize whether a video is played at its natural speed.

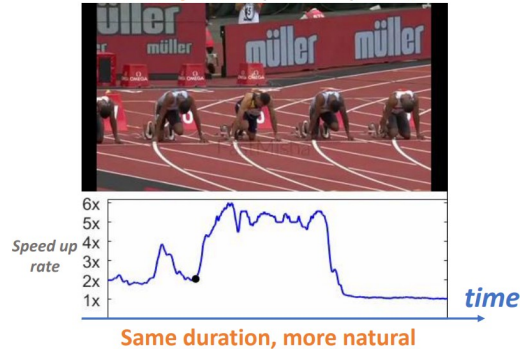
- ▶ **Pretext Task:** A binary classifier predicts whether a clip plays at normal speed or artificially sped up, which forces the network to attend to motion cues.
- ▶ **Applications:** Useful for tasks like action recognition, video summarization, and adaptive video speedup.

Ultimate Goal: Watch video content faster by adaptively speeding up the video

Uniform Speed Up (2x)



Adaptive Speed Up (2x)



[Benaim et al., SpeedNet: Learning the Speediness in Videos, CVPR 2020]

Learning from Videos: **Summary**

- ▶ Video understanding evolves from single-frame analysis to capturing global context using convolution, recurrence, and attention mechanisms.
- ▶ Temporal modeling is essential for tasks like action recognition and future prediction.
- ▶ Self-supervised learning reduces the need for extensive manual annotation.

- ▶ **Computational cost:** 3D convolutions are expensive (C3D: 39.5 GFLOP per clip), and full spatio-temporal attention grows quadratically with sequence length.
- ▶ **Short clips:** Most models see only 16–64 frames at a time; structure spanning minutes still relies on sparse sampling, recurrence, or external memory.
- ▶ **Annotation cost:** Frame-level labels are expensive, and self-supervised pretext tasks do not transfer equally well to every downstream task.
- ▶ **Appearance shortcuts:** Models can classify many actions from scene appearance alone while ignoring motion, which is why motion is evaluated separately.
- ▶ **Open problems:** Efficient architectures for long videos, learning from egocentric and streaming video, and scaling self-supervision to unlabeled video at internet scale.

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